ASSIGNMENT : 12

TITLE : Write a Java program to create and use user-defined packages.

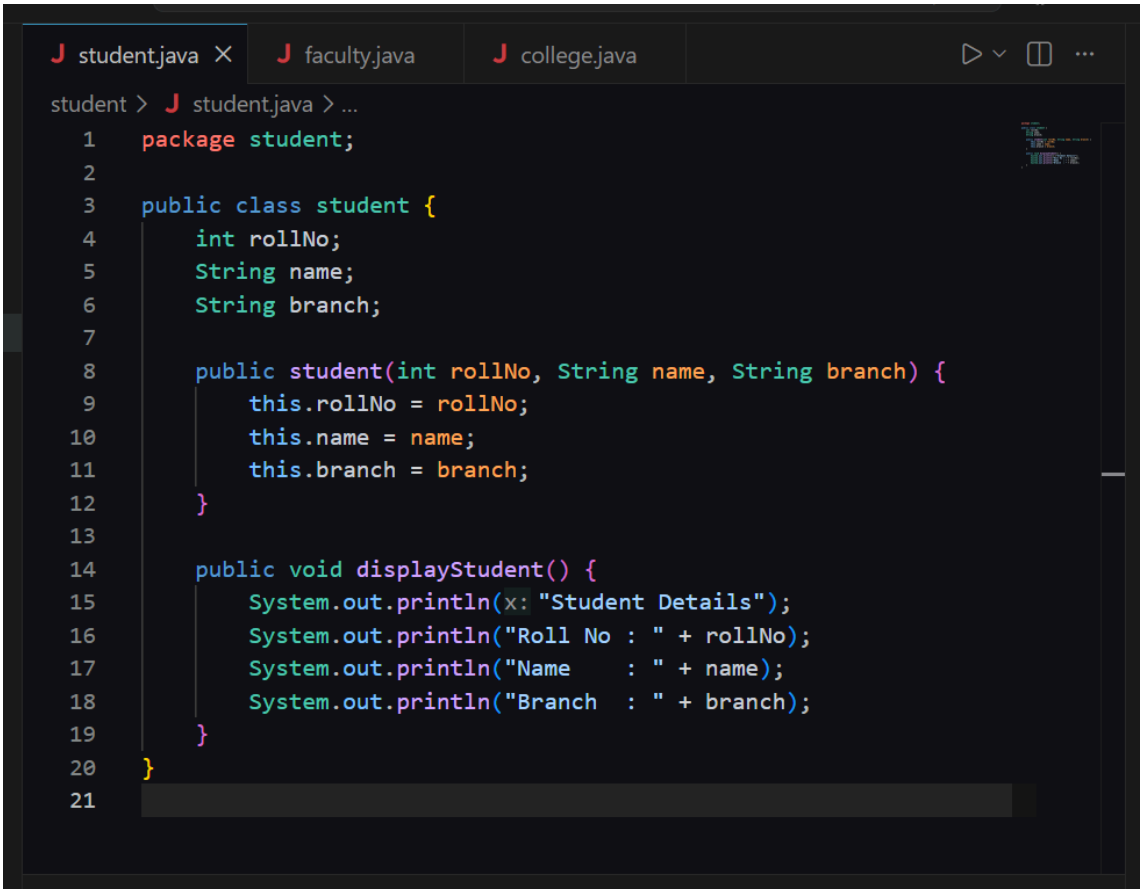
Problem Statement : Write a Java program to create and use user-defined packages.

Learning Objectives : To create and use user-defined packages for organizing Java programs.

Theory : Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

Exercise :

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.



student.java | faculty.java | college.java

college.java > Java > college

```java
import faculty.faculty;
import student.student;

public class college {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {

        student s = new student(rollNo: 101, name: "Anushree", bra
        faculty f = new faculty(facultyId: 201, name: "Anish", sub

        s.displayStudent();
        f.displayFaculty();
    }
}
```

student.java | faculty.java | college.java

faculty > faculty.java > Language Support for Java(TM) by Red Hat > faculty > faculty(int, String,

```java
package faculty;

public class faculty {
    int facultyId;
    String name;
    String subject;

    public faculty(int facultyId, String name, String subject) {
        this.facultyId = facultyId;
        this.name = name;
        this.subject = subject;
    }

    public void displayFaculty() {
        System.out.println(x: "\nFaculty Details");
        System.out.println("Faculty ID : " + facultyId);
        System.out.println("Name       : " + name);
        System.out.println("Subject    : " + subject);
    }
}
```

OUTPUT :

```
                cd "d:\A12 anu\" ; if ($?) { javac college.java } ; if ($?)
{ java college }
Student Details
Roll No : 101
Name    : Anushree
Branch  : Computer Engineering

Faculty Details
Faculty ID : 201
Name       : Anish
Subject    : ENTC
```

2. Create a Java program by developing a **user-defined package** named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

book.java 2 ● | librarymain.class | librarymain.java

library > book.java > Java > book

```java
package library;

public class book {

    int bookId;
    String title;
    String author;
    double price;

    public book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void displayBook() {
        System.out.println(x: "Book Details");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
        System.out.println("Price   : " + price);
    }
}

```

```java
import library.book;

public class librarymain {

    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        book b = new book(bookId: 101, title: "Java Programming", author: "Anushree Shinde", price: 699.00);

        b.displayBook();
    }
}
```

OUTPUT :

```
PS D:\A12b Anu> cd "d:\A12b Anu\" ; if ($?) { javac librarymain.java } ; if ($?) { java librarymain }
Book Details
Book ID : 101
Title   : Java Programming
Author  : Anushree Shinde
Price   : 699.0
PS D:\A12b Anu>
```

Outcome :

- Understood the concept of user-defined packages in Java.
- Created and imported packages to organize related classes.
- Learned how packages improve code reusability, modularity, and maintainability.

Conclusion :

This assignment demonstrated the creation and use of user-defined packages in Java. Packages help organize classes logically, reduce naming conflicts, improve code management, and support modular application development.